# KijaniKiosk End-to-End Delivery

Capstone · Track A: Infrastructure-First

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github.com/ob002/kijani-kiosk-capstone · release v1.0.0

Staging namespace · smoke-gated approval · production deploy · restart alerting

# Problem and scope

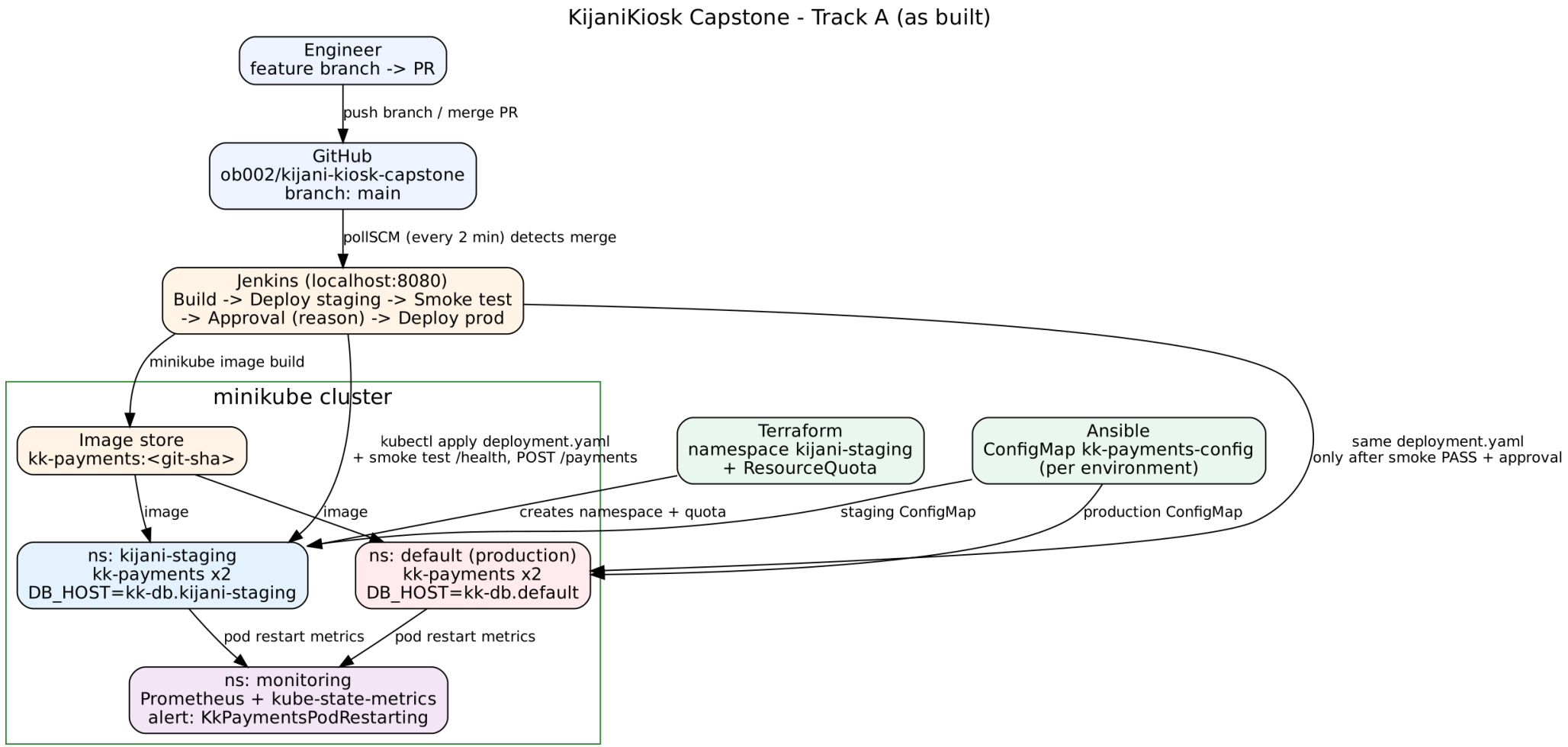
**Problem:** every kk-payments change went straight to production. There was no staging environment, no smoke test, no recorded approval and no alert when the service is unhealthy.

**In scope (built):**

1. `kijani-staging` namespace + ResourceQuota provisioned by **Terraform**
2. Per-environment ConfigMap (different `DB_HOST`) applied by **Ansible**
3. **One** `deployment.yaml` for both environments: probes, CPU/memory limits, JSON logs
4. **Jenkins**: merge to main → staging → smoke test → approval with reason → production
5. **Prometheus** alert on kk-payments pod restarts, committed in `monitoring/`

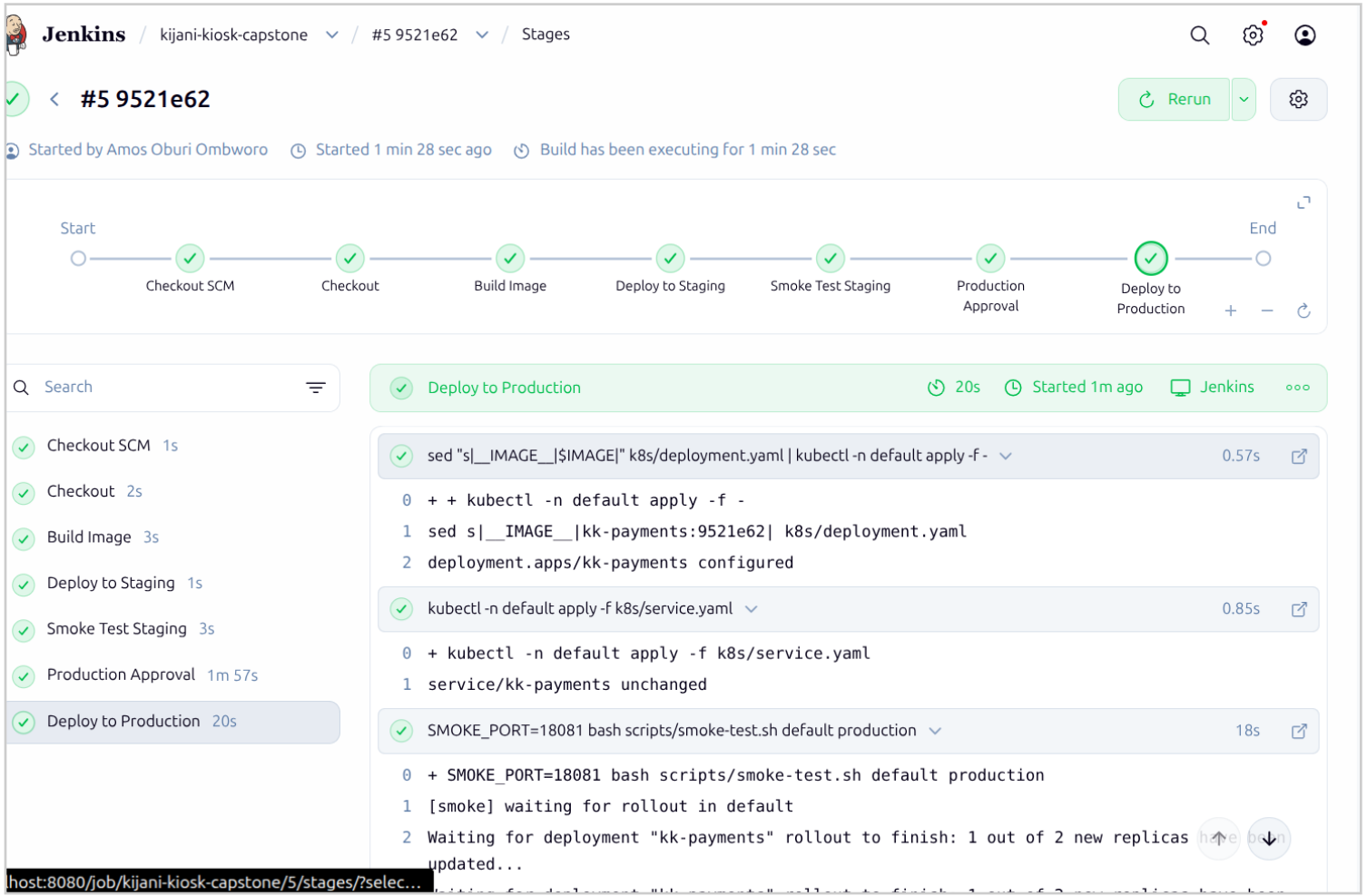
**Out of scope:** separate production cluster (no budget, namespace isolation shows the pattern); Alertmanager routing (needs external accounts).

# Architecture (as built)




Every arrow is labelled with what flows across it. Staging and production run the **same image and manifest**; only the namespace-local ConfigMap differs.

# Pipeline: build #5, end to end




**Build image** tagged with git SHA (`9521e62`)

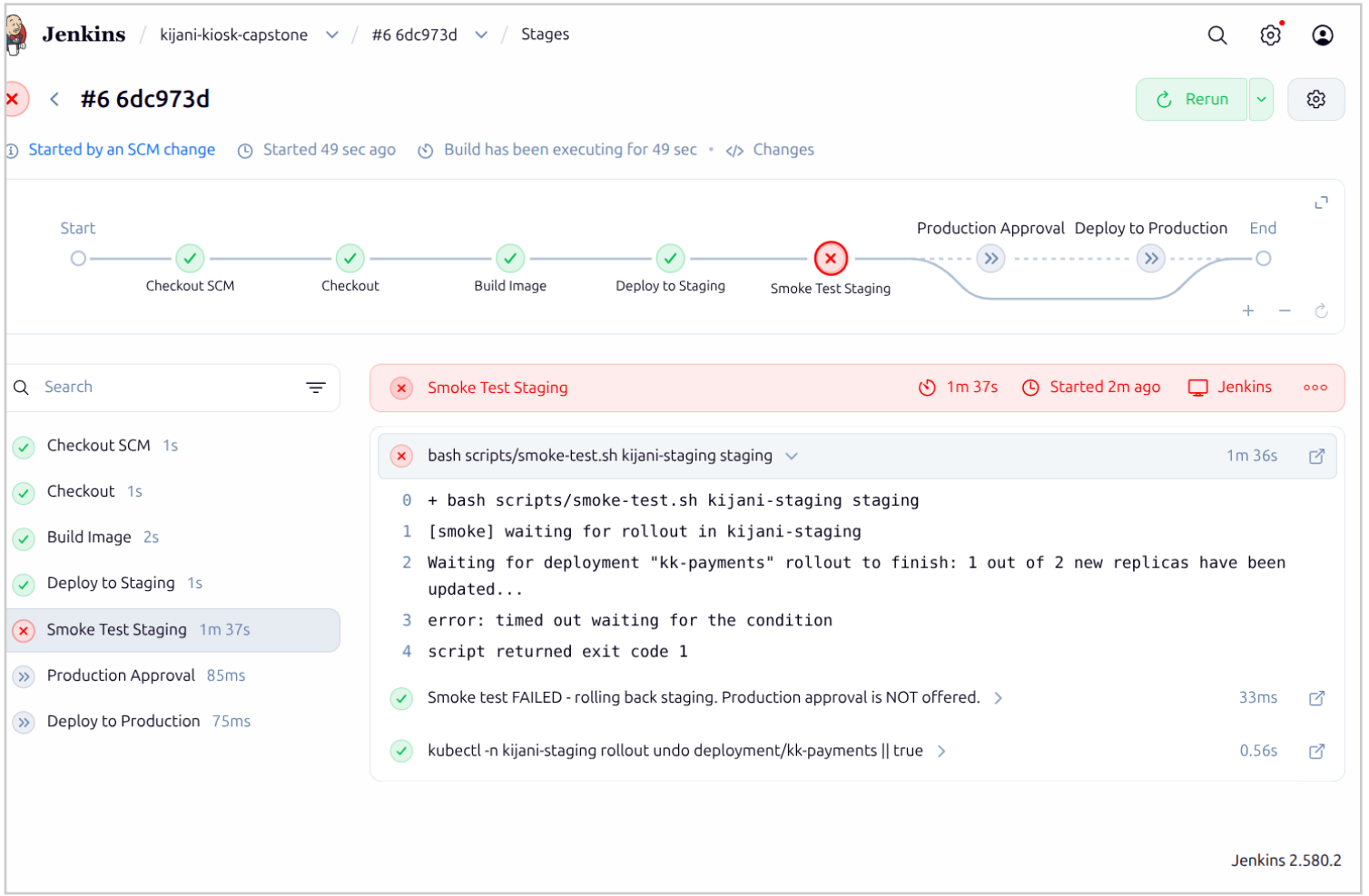
**Deploy to staging** → smoke test PASS (`env=staging`, staging DB_HOST)

**Production Approval**: human gate, reason required and saved in the build description

**Deploy to production** → own smoke test PASS (`env=production`)

Merges trigger the pipeline automatically via `pollSCM`; build #6 says "Started by an SCM change".

# Fault handling: build #6




**Deliberate fault:** PR #3 pointed the health probe at `/broken`.

Merge → pipeline started by itself.

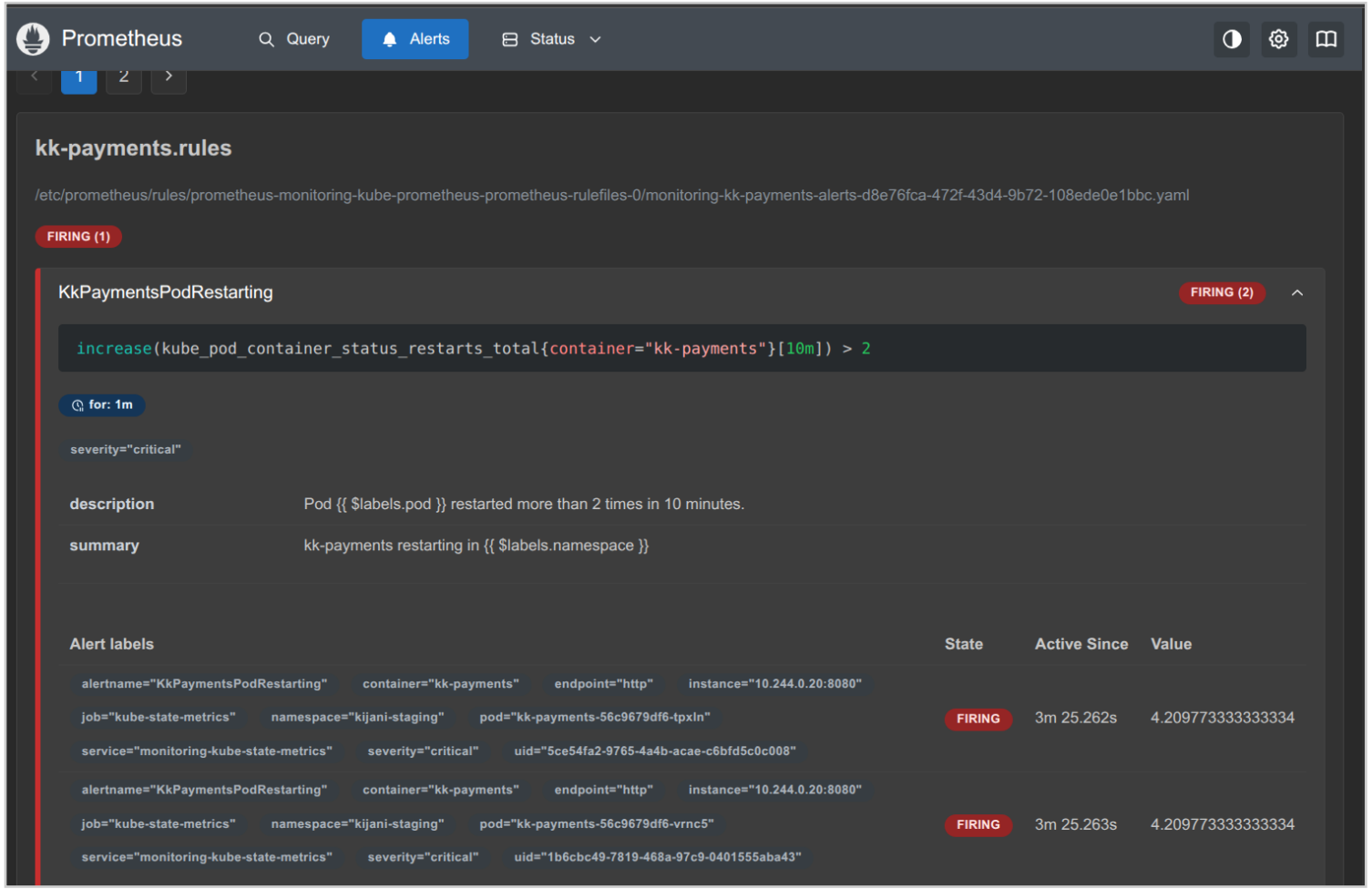
Pods never became Ready, so **Smoke Test Staging failed**.

Jenkins ran `rollout undo` on staging.

**Production Approval and Deploy to Production were skipped**, so production was never touched.

Fixed by PR "fix(k8s): restore health probe path".

# Monitoring: alert fired on a real failure




Rule `KkPaymentsPodRestarting`:

```
increase(kube_pod_container_
status_restarts_total
{container="kk-payments"}[10m]) > 2
```

held for 1 minute

Test: broke the liveness probe in staging, so the pods restarted 5 times.

Result: **FIRING** for both `kijani-staging` pods.

Data source: kube-state-metrics via kube-prometheus-stack.

# Key decision: one manifest, config per namespace

**Decision:** staging and production apply the *identical* `k8s/deployment.yaml`. All differences live in a ConfigMap named `kk-payments-config` that exists separately in each namespace.

**What it buys:** what passed the staging smoke test is byte-for-byte what reaches production. There are no per-environment manifest copies to drift apart.

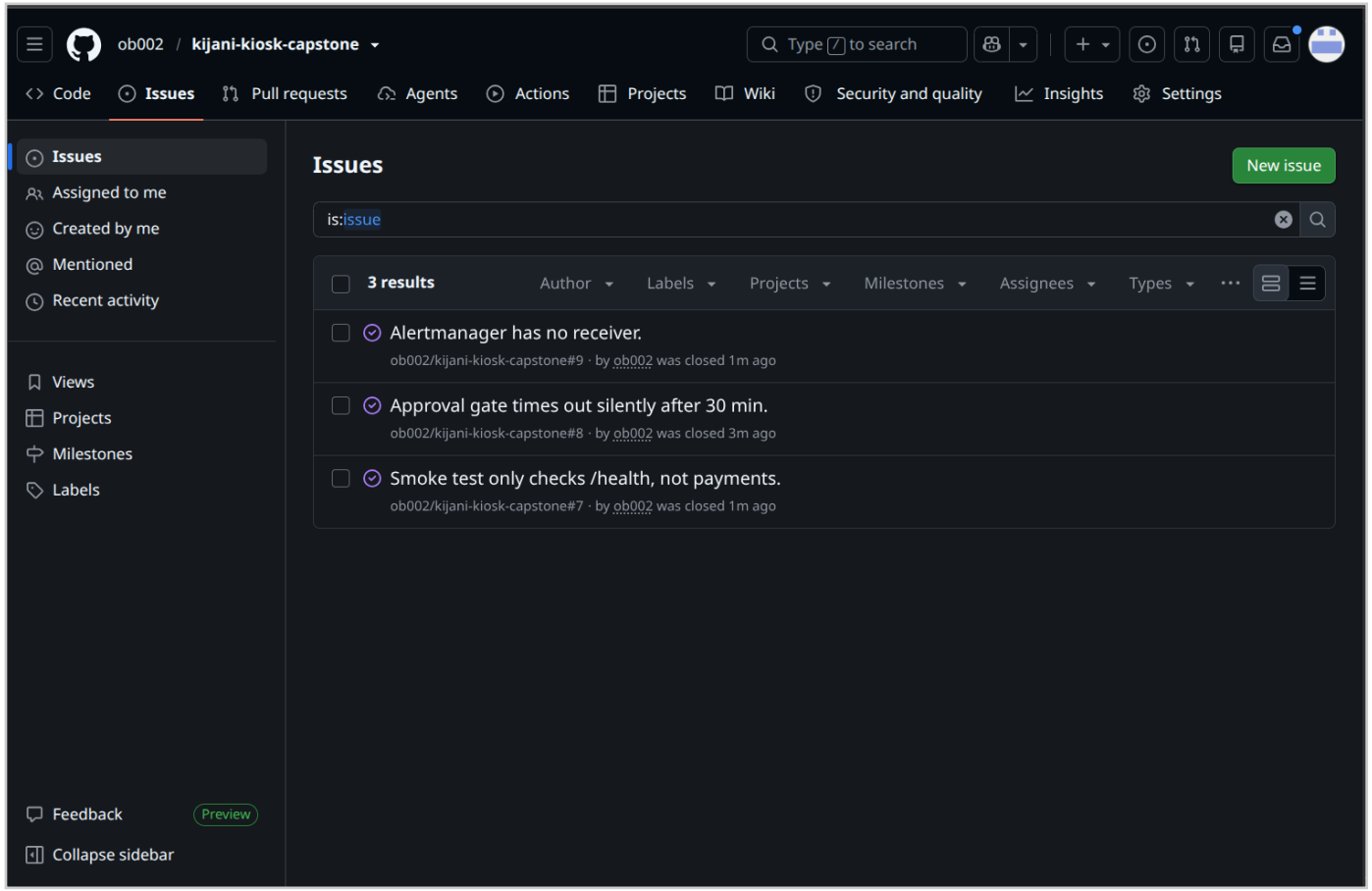
**What it costs:** correctness depends on a *naming convention*. If the ConfigMap is missing or misnamed in one namespace, nothing catches it until runtime. The smoke test checks `env` to cover this.

**Alternative rejected:** Kustomize base + overlays makes environment differences explicit in one diff, but adds a tool and a layer of indirection for a two-environment system.

# AI tooling (matches docs/ai-governance-log.md)

| Entry | AI produced | What it got wrong | My change |
|---|---|---|---|
| 1. Pipeline | Jenkinsfile, smoke-test.sh | Never warned the approval gate times out after 30 min, so build #4 ABORTED. Missed that a stale job held both executors. | Disabled the stale job, cleared the queue, documented the timeout |
| 2. Image build | minikube docker-env + docker build | Assumed a Docker runtime. minikube runs containerd, so the build failed. | Switched to minikube image build, in the Jenkinsfile too |
| 3. Alert test | Alert rule + fault steps | No wait time given; the fault was undone at 0 restarts, so the alert stayed INACTIVE. | Waited until 5 restarts, so it was FIRING |

# Testing and feedback




**#7 Medium:** the smoke test only checked /health. **Fixed** in commit 977862a (PR #11): it now also POSTs /payments and requires 201.

**#8 Low:** the approval gate times out silently. **Resolved** by documenting it in README §5.

**#9 Medium:** Alertmanager has no receiver. **Not planned**: listed as a production gap.

Review type: self-review. (Replace with your peer reviewer's name if a classmate reviewed.)

# Production gaps and remediations

- **Receipt chain not integrated**
  → Staging has RECEIPTS_BUCKET set, but the app does not write to S3 → add S3 PutObject + the Week 10 functions

- **Staging and prod share one minikube cluster**
  → Move production to its own cluster/account

- **No NetworkPolicy**
  → Default-deny ingress per namespace, allow only the monitoring namespace

- **Alerts reach nobody (#9)**
  → Add an Alertmanager Slack/email receiver

- **Images live only inside minikube**
  → Push to a registry (e.g. Nexus/Docker Hub) for multi-node use

- **pollSCM adds up to 2 min delay**
  → GitHub webhook once Jenkins is publicly reachable

- **Secrets/DB not real**
  → Managed DB per environment; secrets via Sealed Secrets